

University of Dhaka
Affiliated Engineering Colleges
B. Sc. in Computer Science & Engineering
2nd Year 1st Semester Final Examination-2021
CSE 2102 (Object Oriented Programming)

Time: 3.00 Hours

Marks: 70

[Answer any Five questions]

1. a) What is OOP? Why it is effective over structured programming? Explain. 3
b) Write down the general form of class. Differentiate between class and objects with example. 3
c) What is meant by data hiding? What are the differences among public, protected and private access specifier in a class? 3
d) Write a program to find out the minimum of two given numbers. The numbers can be integer, long and double. Overload a min() function to handle those three types of data. 5
2. a) A constructor is a special function in a class. Explain why a constructor is different than other member function. Give examples. 3
b) Define constructor overloading? What are the reasons to overload constructor? 3
c) What is interface? Why use java interface? Why multiple inheritance is supported in Interface while it is not supported in case of class? 5
d) Why this keyword is used in java language? 3
3. a) Why should you define your classes within your own package? How can you define a package? 3
b) What are similarities and differences between java class and java interface? 3
c) Define method overriding with suitable example? What are the rules for method overriding? 3
d) Write a simple java program to read data from keyboard and write into file. 5
4. a) What is Byte code in java? 2
b) What do you mean by abstract class? Why it is needed? Explain with example. 6
c) What is Garbage Collection in JAVA? How does Garbage Collection works in JAVA? 6
5. a) What are different access modifiers in Java? 3
b) What is dynamic method dispatch? Illustrated with example. 5
c) Write the definition of package with example. What are the necessities of package? What is the procedure of creating a package? Illustrate with pseudo code. 6
6. a) Discuss how try-catch-finally block works in java. 3
b) Write a program that takes two integers from keyboard to perform division operation. A try block to throw an exception when a wrong type of data is keyed or division by zero is occurred. Also write appropriate catch block to handle the exception thrown. 5
c) How can you define your own exception type in java? Explain with example. 4
d) What happens when a class member is declared as static? What restrictions are applied to a method that is declared as static? 2
7. a) How to check palindrome string in JAVA? 4
b) Is there any wrong or error in the following code? If you find any error then mark, write the correct code & output. 6

i.

```
class Box {
    int width;
    float height;
    double depth;
}
```

ii.

```
class boxDemo {
    public static void main(String args[]) {
        Box mybox = new Box;
        double vol;
        mybox.width = 10.00;
        mybox.height = 20
        mybox.depth = 15.50;
        vol = mybox.width * mybox.height *
        mybox.depth;
        System.out.println("Volume is " + vol); }
```

- c) Write down the output of following code:

i.

```
public class Main {
    public static void main(String[] args)
    {
        int digit = 15;
        digit = 20;
        System.out.println(digit);
    }
}
```

ii.

```
public class Main {
    public static void main(String[] args) {
        int Hello_World= 1111;
        System.out.println("Hello World");
        System.out.println("Welcome To MEC");
        //System.out.println(Hello_World);
        System.out.println("Hello_World"); }
}
```

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